WORKED SOLUTION

STEP III 1987, Question 16

Probability and Statistics • Order Statistics and Poisson

STEP III 1987, Question 16

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/87-s3-q16

The key idea

In part (i), the event "$A \leq k$ and $B \geq 1-k$" is awkward to handle directly, but its complement is a union of two simple events, "all the X_i exceed k" and "all the X_i are below $1-k$". Inclusion–exclusion then does the work. In part (ii), the number of winners is binomial with a huge n and a tiny p, which is exactly where the Poisson distribution lives. So we maximise the Poisson probability of one or two winners, and then make sure the approximation can't push the answer over 60%.

Solution

Part (i)

Let E be the event $A > k$ and F the event $B < 1-k$. Then $A > k$ means every X_i exceeds k, and $B < 1-k$ means every X_i is less than $1-k$. By independence,

$$\mathrm{P}(E) = (1-k)^n, \qquad \mathrm{P}(F) = (1-k)^n, \qquad \mathrm{P}(E \cap F) = (1-2k)^n,$$

where the last one holds because $E \cap F$ says every X_i lies in $(k, 1-k)$, an interval of length $1-2k > 0$.

The event we want is the complement of $E \cup F$, so by inclusion–exclusion

$$p_n = 1 - \mathrm{P}(E \cup F) = 1 - 2(1-k)^n + (1-2k)^n.$$

Since $0 < 1-2k < 1-k < 1$, both powers tend to 0, and so

$$p_n \to 1 \quad \text{as } n \to \infty.$$

Comment. However small we make k, with enough observations we become almost certain that some observation lies within k of 0 and another lies within k of 1. In other words, the sample minimum converges to 0 and the sample maximum to 1 (in probability), so the sample range $B - A$ tends to 1, the length of the whole interval. This is why, for example, A and B are natural estimators of the endpoints of an unknown uniform distribution.

Part (ii)

Let $N = 4\,000\,000$ and let X be the number of winners. Readers win independently, each with probability p, so $X \sim \mathrm{B}(N, p)$. Write $q = 1 - p$. You keep your job exactly when $X = 1$ or $X = 2$, which has probability

$$K(p) = Npq^{N-1} + \binom{N}{2} p^2 q^{N-2}.$$

The Poisson picture. With N huge and p small, X is approximately Poisson with mean $\lambda = Np$, so

$$K(p) \approx h(\lambda), \qquad \text{where } h(\lambda) = \mathrm{e}^{-\lambda}\Big(\lambda + \frac{\lambda^2}{2}\Big).$$

Differentiating,

$$h'(\lambda) = \mathrm{e}^{-\lambda}\Big(1 + \lambda - \lambda - \frac{\lambda^2}{2}\Big) = \mathrm{e}^{-\lambda}\Big(1 - \frac{\lambda^2}{2}\Big),$$

which is positive for $0 < \lambda < \sqrt{2}$ and negative for $\lambda > \sqrt{2}$. So h has its greatest value at $\lambda = \sqrt{2}$:

$$h(\sqrt{2}) = \mathrm{e}^{-\sqrt{2}}(\sqrt{2} + 1) \approx 0.2431 \times 2.4142 \approx 0.5869.$$

Your best choice is $p = \sqrt{2}/N \approx 3.5 \times 10^{-7}$, and even then your chance of survival is only about 58.7%. That is the reasoning the question asks for.

> **Ask yourself** The Poisson distribution is only an approximation. Could the true binomial probability creep above 0.6?

No. We can bound $K(p)$ directly. Since $q \le 1$ and $\binom{N}{2} \le \frac{1}{2}N^2$,

$$K(p) = q^{N-2}\Big(Npq + \binom{N}{2}p^2\Big) \le q^{N-2}\Big(\lambda + \frac{\lambda^2}{2}\Big).$$

Now $1 - p \le \mathrm{e}^{-p}$ for all p (the line $1 - p$ is the tangent to the convex curve e^{-p} at $p = 0$), so $q^{N-2} \le \mathrm{e}^{-(N-2)p} = \mathrm{e}^{-\mu}$, where $\mu = (N-2)p$. Put $c = N/(N-2)$, so that $\lambda = c\mu$ with $c > 1$. Then

$$K(p) \le \mathrm{e}^{-\mu}\Big(c\mu + \frac{c^2\mu^2}{2}\Big) \le c^2\,\mathrm{e}^{-\mu}\Big(\mu + \frac{\mu^2}{2}\Big) = c^2 h(\mu) \le c^2 h(\sqrt{2}).$$

Here $c^2 = (1 - \frac{2}{N})^{-2} < 1.000002$, so for every choice of p,

$$K(p) < 1.000002 \times 0.58694 < 0.587 < 0.6.$$

So however you choose p, you have less than a 60% chance of keeping your job, as required.

Checks

- In part (i), $n = 1$ gives $p_1 = 1 - 2(1-k) + (1-2k) = 0$. That's right: a single observation can't be both $\leq k$ and $\geq 1-k$ when $k < \frac{1}{2}$. For $n = 2$ we get $p_2 = 2k^2$, the probability that one of the two points lands in each end strip, in either order.
- A simulation with $n = 10$, $k = 0.3$ gave 0.944, against the formula's 0.9436.
- Computing the exact binomial $K(p)$ over a fine grid of $\lambda = Np$ gives a maximum of 0.58694 at $\lambda \approx 1.414$, agreeing with the Poisson value to five decimal places.

Common slips

- Writing $p_n = \mathrm{P}(A \leq k)\,\mathrm{P}(B \geq 1-k)$. The minimum and maximum of the same sample are *not* independent.
- Maximising the binomial probability with respect to p directly. It can be done, but the algebra is unpleasant; working with $\lambda = Np$ makes it a one-line calculus problem.
- Stopping at the Poisson approximation without noticing that 0.587 is close enough to 0.6 that some justification is worthwhile.

Beyond STEP

Part (i) is a first taste of *order statistics* and of convergence in probability: the minimum and maximum of a sample converge to the ends of the support. Part (ii) is the Poisson "law of rare events", and the bound we used is a simple instance of the general fact that the binomial and Poisson distributions are very close in total variation when p is small. Both ideas are developed in Grimmett and Welsh, *Probability: An Introduction*, and in Blitzstein and Hwang, *Introduction to Probability*.

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